Subject: ENVIRONMENTAL IMPACT ASSESSMENT

Subject Code: 23A01504c

UNIT – I

Concepts and Methodologies of EIA

I. Introduction to EIA

Environmental Impact Assessment (EIA) is a systematic process of identifying, predicting, evaluating and mitigating the biophysical, social and other relevant effects of a proposed project BEFORE major decisions and commitments are made.

Definition EIA is a planning and decision-making tool. It tells us what the environmental consequences of a development project will be, so that harmful effects can be avoided, reduced or compensated.

Objectives of EIA

- To predict the environmental effects of a proposal early in the planning stage.
- To find ways and means to reduce the adverse effects.
- To shape the project to suit the local environment.
- To present the predictions and options to the decision makers.
- To promote sustainable development and public participation.

Need for EIA

- Rapid industrialisation, urbanisation and mining damage air, water, soil and biodiversity.
- Prevention is cheaper and better than cure (treatment after damage).
- Legal requirement in India under the EIA Notification 2006 (Environment Protection Act 1986).
- Gives transparency in decisions and builds public trust.

2. Initial Environmental Examination (IEE)

IEE is a preliminary, quick and low-cost study done at the early stage of a project to decide whether a full EIA is needed or not.

- **Purpose** screening of the project - to find out if significant impacts are likely.
- It uses existing data and expert judgement; no long field surveys.
- Output of IEE: (a) no further study needed, (b) mitigation measures are enough, or (c) detailed EIA is required.

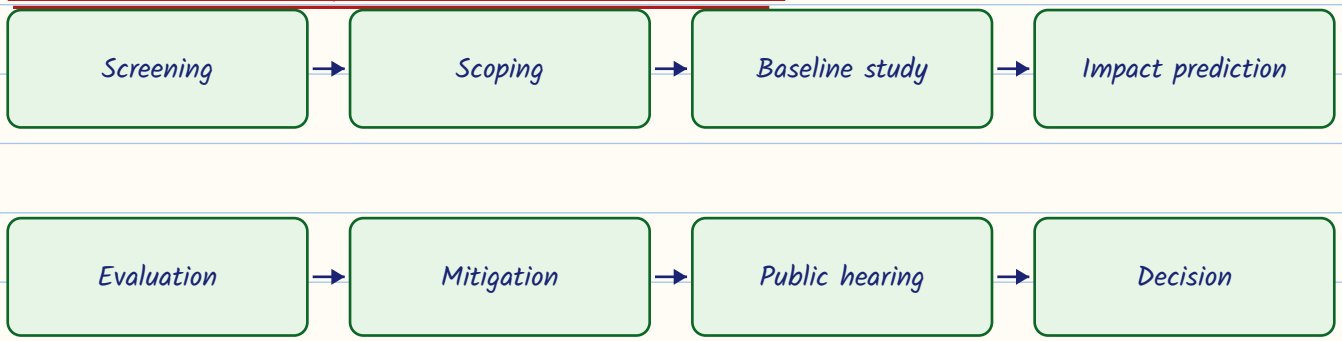
Contents of an IEE report

- Description of the project and its location.
- Description of the existing environment (baseline in brief).
- Likely impacts - positive and negative.
- Possible mitigation measures and alternatives.
- Conclusion and recommendation for full EIA.

IEE vs EIA

Point	IEE	EIA
Depth	Preliminary, brief	Detailed, comprehensive
Time and cost	Less	More
Data	Secondary data mostly	Primary field data and monitoring
Result	Decides need for EIA	Final basis for clearance

3. Elements (Steps) of the EIA Process



- **Screening** decides whether the project needs EIA and at what level (category A or B).
- **Scoping** identifies key issues, alternatives and terms of reference (ToR) for the study.
- **Baseline study** collection of data on the existing air, water, soil, noise, flora, fauna and socio-economic status.
- **Impact identification and prediction** finding what changes the project will cause and how big they are.
- **Impact evaluation** judging the significance of the impacts against standards and public values.
- **Mitigation** measures to avoid, minimise, restore or compensate impacts.
- **Environmental Management Plan (EMP)** who will do what, when and at what cost.
- **Public consultation** hearing the views of affected people.
- **Review and decision** expert committee appraises the report and grants or rejects clearance.
- **Monitoring and follow-up** checking that conditions are followed after the project starts.

4. Factors Affecting EIA (Impact Evaluation and Analysis)

The quality and result of an EIA depend on many factors.

- **Nature and size of the project** a thermal power plant has far more impact than a small workshop.
- **Location and sensitivity of site** near forests, rivers, wetlands, protected areas or dense population increases impact.
- **Quality of baseline data** wrong or old data gives wrong predictions.
- **Prediction methods used** models, experience, simulations.

- **Time scale** short-term, long-term, temporary or permanent impacts.
- **Space scale** local, regional, national or trans-boundary.
- **Reversibility** can the environment recover or not.
- **Public participation** concerns of local people change the final decision.
- **Legal standards and policies** permissible limits for air, water and noise.
- Expertise and funds available to the study team.
- **Alternatives available** other sites, technologies or designs.

Impact evaluation and analysis

Evaluation means judging the importance of a predicted impact. Common criteria are:

- Magnitude (how large) and extent (how wide an area).
- Duration and frequency of the impact.
- Probability (chance of occurring) and reversibility.
- Whether standards or legal limits are crossed.
- Value of the affected resource and number of people affected.
- Cumulative effect with other existing projects.

Impact types direct and indirect; positive and negative; short term and long term; reversible and irreversible; local and regional; cumulative.

5. Preparation of Environmental Base Map

An environmental base map is a map of the study area on which the existing environmental features, sensitive places and the proposed project are marked. It is the starting point for all further analysis.

Steps in preparation

- Select a suitable scale (for example 1:50,000 toposheet of Survey of India).
- Mark the project site and the study area boundary (commonly 10 km radius around the site).

- Show topography, contours, drainage, rivers, lakes and canals.
- Mark land use and land cover – agriculture, forest, settlements, industries.
- Mark roads, railways and other infrastructure.
- **Show sensitive receptors** schools, hospitals, temples, protected areas, wildlife corridors.
- Show sampling locations of air, water, soil and noise monitoring.
- Add north direction, legend, scale and wind rose.

Uses of the base map

- Helps in locating the project properly and in alternative site selection.
- Used for overlaying other maps in the overlay method.
- Helps in understanding spatial spread of impacts.
- Can be prepared using remote sensing and GIS for better accuracy.

6. Classification of Environmental Parameters

Environmental parameters are the measurable features of the environment that may be changed by a project. They are classified into four groups.

Category	Parameters (examples)
Physical / Chemical	Air quality, water quality, soil quality, noise, climate, geology, hydrology
Biological	Flora, fauna, aquatic life, forests, endangered species, habitats
Socio-economic	Population, employment, health, income, housing, transport, education
Cultural / Aesthetic	Historical sites, monuments, religious places, landscape beauty, recreation

- Each parameter is described by indicators, for example BOD and DO for water, SO2 and NOx for air, dB(A) for noise.
- Parameters are given importance weights in some methods (like EQI and

matrix).

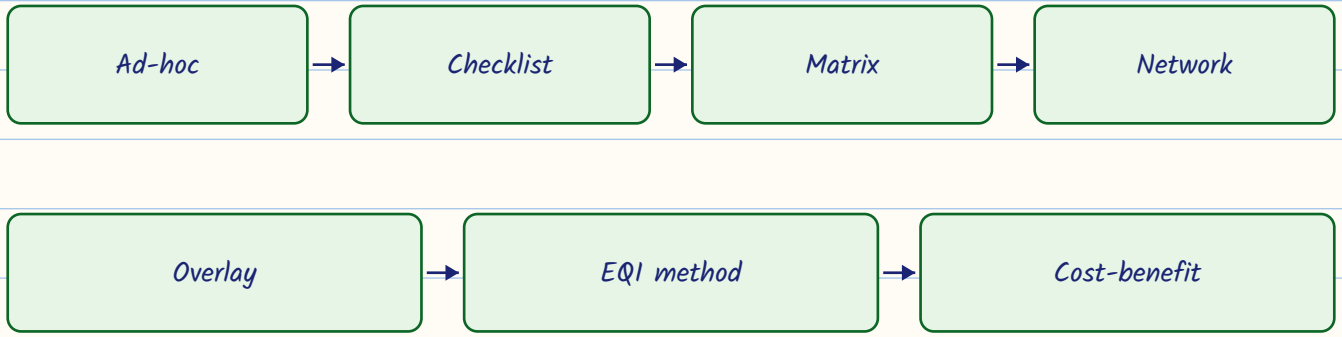
7. Criteria for Selection of EIA Methodology

No single method suits all projects. The choice depends on the following criteria.

- Type, size and complexity of the project.
- Type and quantity of data available.
- Stage of planning (early or detailed).
- Time, money and skilled manpower available.
- Ability of method to identify and compare alternatives.
- Ability to handle direct, indirect and cumulative impacts.
- Ability to give quantitative results.
- Whether it can be understood by the public and decision makers.
- Requirement of the law or the funding agency.
- Flexibility and reproducibility of the method.

8. EIA Methods

EIA methods are tools used to identify, predict and evaluate impacts. The main ones are:



8.1 Ad-hoc Methods

- These give minimum guidance; experts simply list the broad areas of impact based on their experience.

- Impacts are described in general terms - no fixed format.
- **Advantages** quick, cheap, useful for small projects and for a first look.
- **Disadvantages** no uniformity, depends on the experts, can miss important impacts, subjective.

8.2 Checklist Methods (related)

- A list of environmental factors to be checked against the project activities.
- **Types** simple (list only), descriptive (with notes on measurement and prediction), scaling (with ratings) and scaling-weighting checklists.
- Simple to use and good for screening, but do not show interaction among impacts.

8.3 Matrix Methods

A matrix shows project activities along one axis and environmental factors along the other. A mark is placed in a cell where an activity affects a factor.

Leopold Matrix (1971) developed by Luna Leopold of the US Geological Survey. It has 100 project actions (columns) and 88 environmental characteristics (rows), giving 8800 possible interactions.

- In each cell a slash is drawn when an impact is possible.
- Then two numbers from 1 to 10 are written: MAGNITUDE (top left, how big - with + or - sign) and IMPORTANCE (bottom right, how significant).
- Example cell: 8 over 5 means large magnitude, medium importance.
- **Advantages** covers many actions and factors, shows direct link between cause and effect, easy to present.
- **Disadvantages** does not show secondary (indirect) impacts, does not consider time and space, values are subjective, very large for big projects.

Project activity	Air	Water	Soil	Noise	Jobs
Land clearing	-	-	-	-	+
Construction	-	-	-	-	+

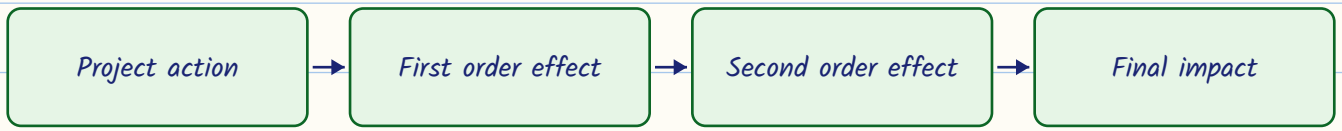
Operation	-	-	-	-	+

Other matrix types

- **Simple interaction matrix** only marks presence or absence of impact.
- **Weighted (quantitative) matrix** gives numerical values and totals them.
- **Stepped (cross-impact) matrix** tries to show second and third order effects.

9. Network Method

- Network (flow diagram) methods link the causes and effects as a chain, so that direct and indirect impacts can be seen together.
- Developed to overcome the drawback of matrices which show only first order effects.
- **Sorensen network (1971)** made for coastal zone planning. It starts from land use activities, then shows the causal factors, the environmental changes and finally the impacts on people.



- **Example** Deforestation for a road -> soil erosion -> siltation of river -> loss of fish -> loss of income of fishermen.
- **Advantages** shows indirect and cumulative impacts, good for understanding cause-effect pathways.
- **Disadvantages** becomes complex and untidy for large projects, mostly qualitative, no ranking of importance.

10. Environmental Media Quality Index Method

- Also called the Environmental Quality Index (EQI) or Battelle Environmental Evaluation System (EES) method, developed by Battelle Columbus Laboratories

(1972) for water resources projects.

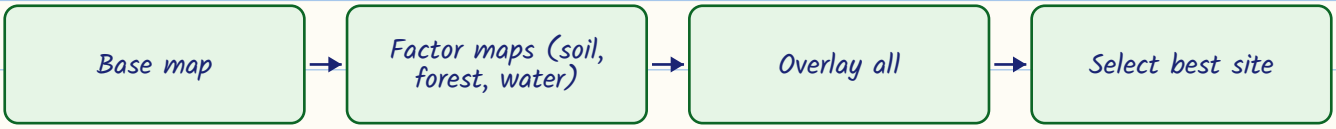
- **Environment is divided into 4 categories** Ecology, Environmental Pollution, Aesthetics and Human Interest.
- These have 18 components and 78 parameters.
- A total of 1000 Parameter Importance Units (PIU) are shared among all parameters (weights).
- For each parameter, the measured value is converted into an Environmental Quality (EQ) scale from 0 (very bad) to 1 (very good) using a value function curve.

Formula Environmental Impact Units (EIU) = sum of [PIU x EQ]. The index is calculated for "with project" and "without project" conditions; the difference is the net impact.

- **Advantages** gives numerical and comparable results, weights show relative importance, good for comparing alternatives.
- **Disadvantages** costly and time consuming, value functions are subjective, needs a lot of data.

II. Overlay Methods

- Overlay method uses a set of transparent maps (mylar sheets), one for each environmental factor.
- Each map is shaded darker where the sensitivity or impact is higher.
- When all the maps are placed over one another on the base map, the combined darkness shows areas of high total impact and low impact.



- **McHarg's overlay method (1969)** Ian McHarg used it for highway route

selection, choosing the path with the least total environmental cost.

- Today the same idea is done with GIS using digital layers (GIS overlay analysis).
- **Advantages** easy to understand, good for site and route selection, spatial picture of impact.
- **Disadvantages** all factors get equal importance unless weighted, cannot show indirect or time based impacts, limited number of sheets can be read clearly.

12. Cost / Benefit Analysis

- A method of comparing the total costs of a project with its total benefits, both expressed in money terms.
- Includes environmental costs (loss of forest, health damage) and environmental benefits (clean energy, flood control) along with economic ones.
- **Benefit Cost Ratio** BCR = Present value of benefits / Present value of costs. A project is acceptable if BCR is greater than 1.
- **Net Present Value** NPV = PV of benefits - PV of costs; must be positive.
- Future values are brought to present using a discount rate: $PV = F / (1 + r)$ raised to n.

Steps

- Define the project and alternatives.
- Identify all costs and benefits, direct and indirect.
- Give money values (market price, replacement cost, willingness to pay).
- Discount to present value and compare.
- Do sensitivity analysis for different discount rates.
- **Advantages** simple decision rule, shows economic efficiency.
- **Disadvantages** difficult to put a price on life, health, beauty and species; discount rate favours present generation; ignores who gains and who loses.

13. Comparison of Methods (Quick Revision)

Method	Key feature	Main limitation
Ad-hoc	Expert opinion, quick	Subjective
Checklist	List of factors	No interaction shown
Matrix	Action vs factor grid	No indirect impact
Network	Cause-effect chain	Complex, qualitative
EQI / Battelle	Numerical weighted index	Costly, data heavy
Overlay	Map layers	Equal weight, no time
Cost-benefit	Money terms	Pricing of environment

Important questions (1) Define EIA and explain its elements. (2) Differentiate IEE and EIA. (3) Explain the Leopold matrix. (4) Describe the EQI method. (5) Write short notes on overlay and network methods. (6) Criteria for selecting an EIA method.

14. Worked Illustration – Leopold Matrix

Suppose a cement factory is proposed near a village. Land clearing is Action A, Quarrying is Action B and Stack emission is Action C. Environmental factors are air quality, groundwater and employment.

Factor	A Clearing	B Quarrying	C Stack emission
Air quality	-3 / 4	-6 / 7	-9 / 9
Groundwater	-2 / 3	-7 / 8	-1 / 2
Employment	+5 / 6	+7 / 7	+4 / 5

- Each entry is Magnitude / Importance. A minus sign shows adverse impact and a plus sign shows beneficial impact.
- Add the cells row-wise to see which factor is hit most and column-wise to see which action is the worst.
- Here stack emission is the most harmful for air quality; so the main mitigation

is an air pollution control system such as an ESP and a tall stack.

Remember the matrix gives only a first look. It must be followed by detailed prediction using models and field data.

15. Environmental Management Plan (EMP) in Brief

- EMP is the document that turns EIA findings into action.
- It lists mitigation measures, responsible persons, schedule, budget and monitoring programme.
- **Parts** air pollution control plan, water and effluent plan, solid waste plan, green belt plan, noise control, occupational health and safety, disaster management and rehabilitation plan.
- A green belt around the project reduces dust and noise and improves appearance.
- Regular monitoring reports are submitted to the pollution control board and the ministry.

16. Key Points to Remember

- EIA is preventive; it is done before the project starts.
- IEE is the quick study that decides whether full EIA is required.
- Baseline data are collected for at least one season, normally three months for air and water.
- Ad-hoc is the simplest method; EQI is the most quantitative; overlay is best for spatial planning; network is best for indirect impacts.
- Leopold matrix = 100 actions x 88 factors; Battelle system = 1000 PIU, 78 parameters.
- Cost-benefit analysis is acceptable when BCR > 1.

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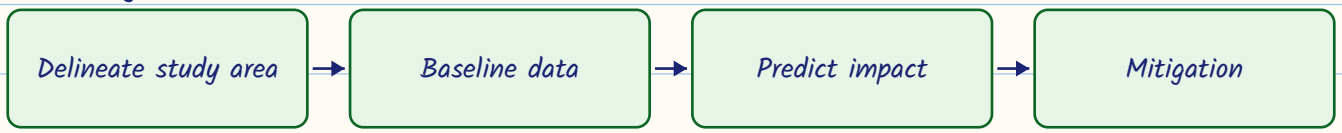
UNIT – II

Impact of Developmental Activities and Land Use

1. Introduction

Development activities such as mining, industries, dams, highways, irrigation, housing and power plants change the land, water, air and living things around them. EIA studies each component of the environment separately and then together.

- **Components studied in this unit** soil, groundwater, surface water, air and biological environment.
- For each component the general method is the same: define study area -> collect baseline data -> predict impact -> judge significance -> suggest mitigation.



Land use impact when forest or farm land is changed to industry or township, the soil structure, drainage, groundwater recharge, local climate and habitat all change.

2. Assessment of Impact on Soil and Groundwater

2.1 Introduction

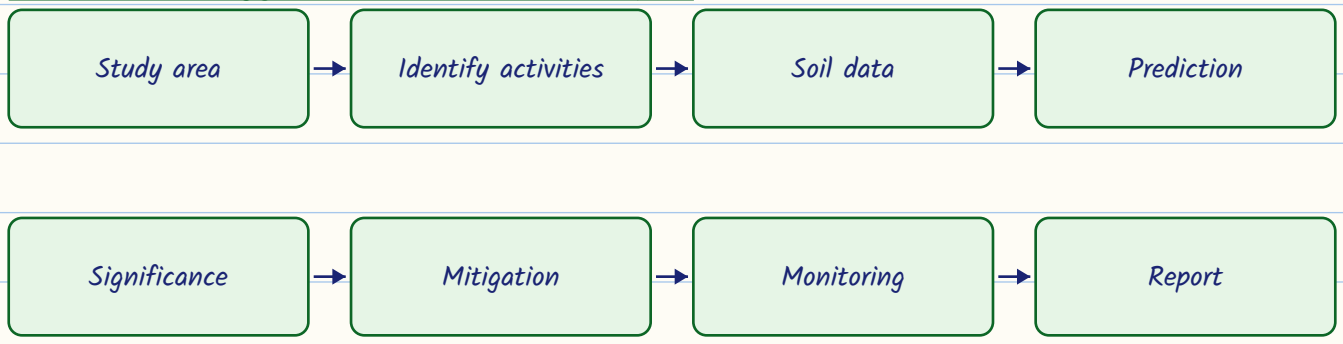
- Soil is a non-renewable resource in the human time scale; it takes hundreds of years to form a few centimetres.
- Soil supports crops, forests and filters water going down to the groundwater.

- Groundwater is the water stored in aquifers below the ground; it is used for drinking, irrigation and industry.
- Developmental activities cause soil erosion, compaction, loss of fertility, salinity, contamination and fall in groundwater table.

Common sources of impact

- **Mining and quarrying** removal of top soil, overburden dumps, acid mine drainage, lowering of water table.
- **Industries** effluent discharge, hazardous waste dumping, leakage of storage tanks.
- **Agriculture** fertilisers, pesticides, over irrigation causing waterlogging and salinity.
- **Construction** excavation, compaction, loss of recharge area due to paving.
- **Landfills** leachate entering aquifers.
- **Over pumping of borewells** depletion and seawater intrusion in coastal zones.

2.2 Methodology for Assessment (steps)



3. Delineation of the Study Area

Study area delineation means fixing the boundary of the area that may be affected by the project.

- Usually a radius of 5 to 10 km around the project site is taken (core zone is the project site and buffer zone is the 10 km radius).

- **Boundary is based on** topography, drainage pattern, groundwater flow direction, wind direction, location of villages, forests and water bodies.
- For groundwater, the catchment or the aquifer boundary is more important than a circle.
- The area is shown on the base map with the sampling stations.
- Larger area is taken for projects with larger emission or discharge, such as thermal power plants.

4. Identification of Activities

All project activities that can affect soil and groundwater are listed phase wise.

Phase	Activities	Possible effect
Pre-construction	Site survey, land acquisition, clearing	Loss of vegetation, soil exposure
Construction	Excavation, filling, transport, camps	Erosion, compaction, oil spills
Operation	Effluent, solid waste, fuel storage, water drawl	Contamination, groundwater fall
Closure / decommissioning	Dismantling, waste removal	Residual contamination

- **Activities are grouped as** land use change, material handling, waste disposal, water abstraction and accidental releases.

5. Procurement of Relevant Soil Quality Data

Baseline soil data are collected by field sampling and from secondary sources.

Secondary data

- Soil maps and reports from NBSS&LUP, State agriculture department and Survey of India toposheets.
- Land use data from remote sensing (NRSC, satellite images).
- Rainfall, geology and groundwater data from IMD, GSI and CGWB.

Primary data – soil sampling

- Samples are taken from different locations (agricultural land, forest, barren land, near the site and in the buffer zone).
- **Depth of sampling** usually 0 to 30 cm (surface); sometimes up to 90 cm in layers for profile study.
- **Quartering method** soil is collected from several points, mixed, heaped, divided into four parts and two opposite parts are rejected until about 500 g remains.
- Samples are air dried, powdered, sieved (2 mm) and sent to laboratory.

Parameters analysed

Type	Parameters
Physical	Texture, bulk density, porosity, water holding capacity, infiltration rate
Chemical	pH, electrical conductivity, organic carbon, N, P, K, cation exchange capacity, sodium adsorption ratio
Contaminants	Heavy metals (Pb, Cd, Cr, Hg), oil and grease, pesticides

Groundwater data

- **Collected from open wells and borewells** depth to water level (pre and post monsoon), pH, TDS, hardness, chloride, fluoride, nitrate, heavy metals and bacteria.
- Standards used: IS 10500 for drinking water.

6. Impact Prediction

Prediction means estimating the changes in soil and groundwater quality because of the project.

- **Soil erosion** predicted using the Universal Soil Loss Equation (USLE).

USLE $A = R \times K \times LS \times C \times P$ where A = soil loss (tonnes per hectare per year), R = rainfall erosivity, K = soil erodibility, LS = slope length and steepness factor, C = crop or cover factor, P = conservation practice factor.

- **Contamination of soil** mass balance method (input of pollutants - output = accumulation) and dispersion of air pollutants that settle on soil.
- **Groundwater level** Darcy's law for flow, $Q = K \times i \times A$, and groundwater flow or numerical models (MODFLOW).
- **Groundwater quality** contaminant transport models predict movement of leachate or effluent plume.
- **Analogy method** comparing with the results observed at a similar existing project.
- Expert opinion and checklists are used where data is not enough.
- Prediction is done for different stages and for worst case conditions.

7. Assessment of Impact Significance

After prediction, the predicted change is compared with a standard or a threshold to say whether the impact is important.

- **Compare predicted values with** IS 10500, CPCB standards, FAO soil guidelines and background (baseline) levels.
- **Consider** magnitude, extent, duration, reversibility, sensitivity of receptor (farm land, drinking water source) and number of people affected.
- **Rating given as** negligible, minor, moderate, major or severe.
- **Significant impact examples** pollutant level exceeding permissible limit, loss of prime agricultural land, fall of water table beyond natural fluctuation, contamination of a drinking water source.
- Cumulative impacts of nearby industries are also considered.

8. Identification and Incorporation of Mitigation Measures

Mitigation follows the hierarchy: Avoid -> Minimise -> Restore -> Compensate.

For soil

- Preserve and stockpile the top soil separately and reuse it for green belt and reclamation.
- Terracing, contour bunding, check dams and grassing on slopes to control erosion.
- Proper design of waste dumps with retaining walls and garland drains.
- Lined and covered storage for hazardous waste; engineered landfills with leachate collection.
- **Soil remediation methods** bioremediation, phytoremediation, soil washing and excavation.
- Afforestation and green belt with local species.

For groundwater

- Rain water harvesting and artificial recharge pits to compensate for abstraction.
- Impermeable lining of ponds, tanks and drains.
- Treatment of effluent to standards before use or disposal; zero liquid discharge where possible.
- Controlled pumping and use of recycled water.
- Monitoring wells around the site for regular quality checks.
- These measures are written into the Environmental Management Plan with cost and responsibility.

9. EIA in Surface Water Environment

9.1 Introduction

Surface water includes rivers, streams, lakes, ponds, reservoirs, estuaries and the sea. Projects affect it by discharging wastewater, changing flow, increasing siltation and altering drainage.

9.2 Sources of impact

- Industrial effluents and sewage discharge.
- Thermal discharge from power plants (thermal pollution).
- Runoff from construction sites, mines and agricultural fields (nutrients, pesticides, sediments).
- Dams and barrages changing flow, sediment movement and fish migration.
- Dredging, reclamation and port activities.
- Solid waste and oil spills.

9.3 Methodology for assessment

- **Step 1** Describe the project and list the water related activities (water drawl, effluent, runoff).
- **Step 2** Delineate the study area - the catchment, the stretch of river and downstream users.
- **Step 3** Baseline data collection - flow, depth, velocity, rainfall, and water quality.
- **Step 4** Identify water uses (drinking, irrigation, fishing, bathing, industry) and the water quality criteria for each use.
- **Step 5** Predict impacts by modelling.
- **Step 6** Evaluate significance against standards.
- **Step 7** Suggest mitigation and monitoring.

Baseline parameters

Group	Parameters
Physical	Temperature, colour, turbidity, TSS, TDS

Chemical	pH, DO, BOD, COD, nitrates, phosphates, chlorides, oil and grease, heavy metals
Biological	Coliform count, plankton, benthic organisms, fish

Prediction models

- **Dissolved oxygen sag model (Streeter-Phelps)** predicts the fall of DO downstream of a waste discharge and the critical point where DO is minimum.

Streeter-Phelps $dD/dt = K1 \times L - K2 \times D$ where D is the oxygen deficit, L is the BOD remaining, K1 is the deoxygenation rate and K2 is the reaeration rate.

- **Dilution and mixing model** $C = (Qr \times Cr + Qw \times Cw) / (Qr + Qw)$ gives the concentration after complete mixing of the river and the waste.
- Lake and reservoir models for eutrophication; hydrodynamic models for estuaries.

Mitigation

- Treat effluent (primary, secondary and tertiary) before discharge; reuse and recycle.
- Cooling towers or ponds to reduce thermal pollution.
- Silt traps and sedimentation basins at construction sites.
- Maintain minimum environmental flow downstream of dams.
- Buffer strip of vegetation along river banks.
- Regular monitoring of water quality and emergency plan for spills.

10. EIA in the Air Environment

10.1 Air pollution sources

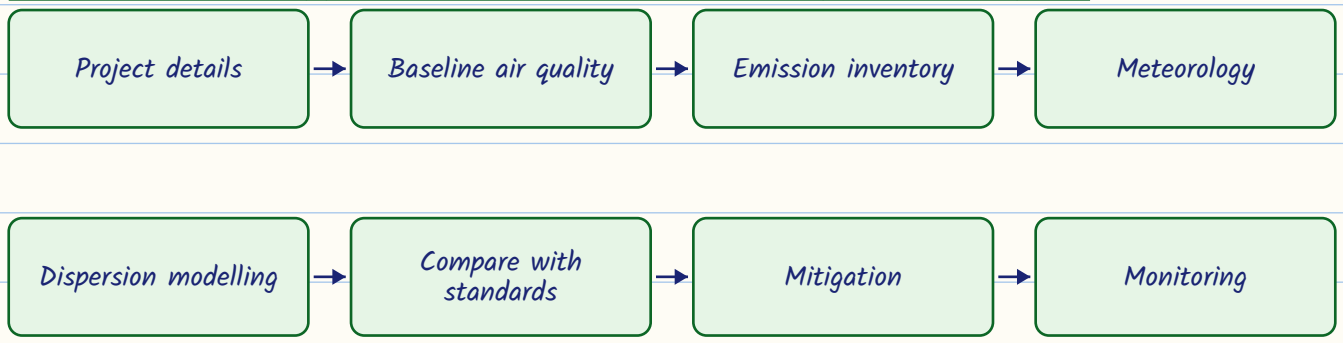
- **Point sources** chimneys and stacks of industries and power plants.
- **Line sources** roads, highways and railway lines (vehicle exhaust).
- **Area sources** open burning, mining areas, construction sites, landfills, towns.

- **Natural sources** dust storms, volcanoes, forest fires.
- **Primary pollutants** SO2, NOx, CO, SPM, PM10, PM2.5, hydrocarbons, lead.
- **Secondary pollutants** ozone, smog, acid rain formed by reactions in the atmosphere.

Effects

- **On health** respiratory diseases, heart diseases, irritation to eyes.
- **On plants** leaf damage, reduced yield. On materials :: corrosion of buildings and monuments (e.g. Taj Mahal).
- **On climate** global warming, ozone layer damage.

10.2 Generalised approach for assessment of air pollution impact



- **Step 1** Collect project details - fuel used, production, stack height, emission rate.
- **Step 2** Baseline air quality monitoring at a number of stations (upwind, downwind and in villages) for SPM, PM10, PM2.5, SO2, NOx, CO as per CPCB; monitoring for 24 hours twice a week for a season.
- **Step 3** Meteorological data - wind speed and direction, temperature, humidity, rainfall, stability class (from IMD or site station); wind rose is drawn.
- **Step 4** Emission inventory - list all sources and their emission rates.
- **Step 5** Dispersion modelling to predict ground level concentration (GLC).
- **Step 6** Add predicted concentration to baseline and compare with National Ambient Air Quality Standards (NAAQS).

- **Step 7** Plan control measures and monitoring.

Dispersion models

- **Gaussian plume model** the basic model, assumes pollutant spreads like a normal (bell shaped) curve horizontally and vertically.

Gaussian plume C(x,y,z) = [Q / (2 x pi x u x sigma-y x sigma-z)] x exp(-y squared / 2 sigma-y squared) x [exp(-(z-H) squared / 2 sigma-z squared) + exp(-(z+H) squared / 2 sigma-z squared)]

- Here Q = emission rate (g/s), u = wind speed (m/s), H = effective stack height, sigma-y and sigma-z = dispersion coefficients (depend on stability class and distance x).
- **Software commonly used** ISCST3, AERMOD, CALPUFF and the line source model CALINE for roads.
- Effective stack height H = physical stack height + plume rise.

Mitigation for air

- **Control equipment** ESP, bag filter, cyclone, scrubber for particles; flue gas desulphurisation for SO2.
- Tall stacks for dispersion; use of clean fuels (gas instead of coal).
- Dust suppression by water sprinkling and covered conveyors.
- Green belt, good housekeeping, vehicle maintenance.
- Continuous emission monitoring system (CEMS).

II. EIA in the Biological Environment (Overview)

- Biological environment means flora, fauna and ecosystems of the study area.
- **Baseline** species list, density, diversity indices (Shannon-Wiener), endangered species, forests, protected areas, migration routes.
- **Methods** quadrat and transect survey, line transect for animals, secondary data from Forest department and Zoological and Botanical Surveys.

- **Impacts** loss of habitat, fragmentation, mortality, pollution effects and invasive species.
- **Mitigation** avoid sensitive areas, compensatory afforestation, wildlife corridors, translocation and conservation plan.
- More detail is given in Unit III.

12. Quick Revision Table

Component	Key parameters	Prediction tool	Typical mitigation
Soil	pH, EC, organic carbon, metals	USLE, mass balance	Top soil conservation, terracing
Groundwater	Water level, TDS, hardness	Darcy law, MODFLOW	Recharge, lining
Surface water	DO, BOD, COD, turbidity	Streeter-Phelps, mixing model	ETP, silt traps
Air	SPM, SO2, NOx, CO	Gaussian plume, AERMOD	ESP, stack, green belt

Important questions (1) Explain the methodology for assessment of impact on soil and groundwater. (2) Describe delineation of study area. (3) Explain the steps in assessing impacts on surface water. (4) Explain the generalised approach for air pollution impact assessment. (5) Write about Gaussian plume model and mitigation of air pollution.

13. Worked Examples

Example 1 - Mixing of waste with river water

A river has flow Qr = 10 m3/s with BOD Cr = 2 mg/L. A factory discharges Qw = 0.5 m3/s of waste with BOD Cw = 200 mg/L. Find the BOD after complete mixing.

- C = (Qr x Cr + Qw x Cw) / (Qr + Qw)
- C = (10 x 2 + 0.5 x 200) / (10 + 0.5)

- C = (20 + 100) / 10.5 = 11.43 mg/L

Result BOD after mixing is about 11.4 mg/L, which is much higher than the river's original 2 mg/L. So treatment of the waste is needed.

Example 2 – Soil loss by USLE

Given R = 400, K = 0.3, LS = 1.2, C = 0.5, P = 0.8.

- A = R x K x LS x C x P
- A = 400 x 0.3 x 1.2 x 0.5 x 0.8 = 57.6 tonnes per hectare per year
- Comparing with the tolerable soil loss (about 10 to 12 t/ha/year in India), the loss is high, so conservation practices such as contour bunding and cover crops are needed.

14. Key Points to Remember

- Study area is normally a 10 km radius, but groundwater uses the aquifer or catchment boundary.
- Soil sampling is done at 0 to 30 cm depth and by the quartering method.
- **Prediction tools** USLE for erosion, Darcy and MODFLOW for groundwater, Streeter-Phelps for river DO, Gaussian plume for air.
- Significance is judged against standards (IS 10500, CPCB, NAAQS) and baseline.
- **Mitigation hierarchy** avoid, minimise, restore, compensate.
- Baseline air monitoring is done for one season, twice a week, for 24 hours.

Subject: ENVIRONMENTAL IMPACT ASSESSMENT

Subject Code: 23A01504c

UNIT - III

Assessment of Impact on Vegetation, Wildlife and Risk Assessment

1. Introduction

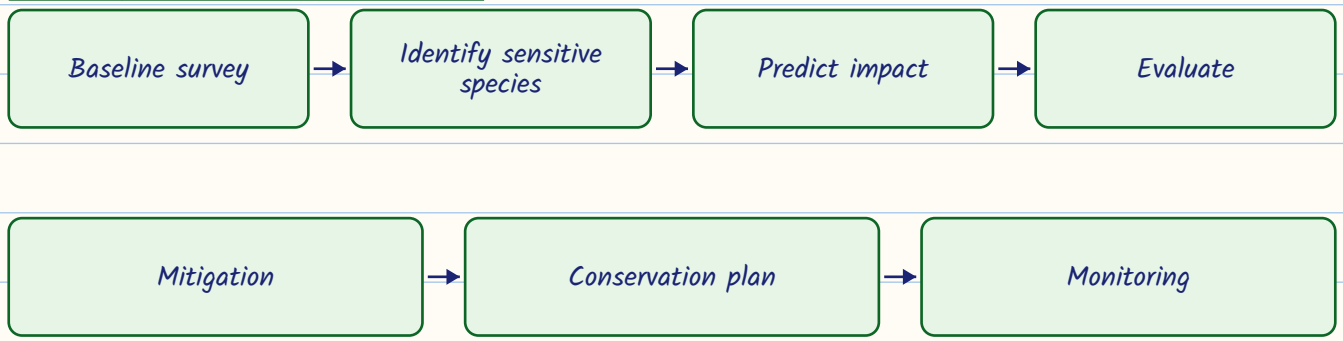
Vegetation (flora) and wildlife (fauna) form the biological environment. They give us food, timber, medicine, clean air, climate regulation and beauty. Developmental projects destroy or disturb them, so their assessment is a very important part of EIA.

- **Biodiversity** the variety of life at gene, species and ecosystem levels.
- India is one of the 17 mega-diverse countries and has 4 biodiversity hotspots (Himalaya, Western Ghats, Indo-Burma, Sundaland - Nicobar).
- Biological components are good indicators of the health of the environment, because plants and animals respond to pollution and habitat change.

Aim of the biological assessment to know what species are present, how sensitive they are, how the project will affect them and how the loss can be avoided, reduced or compensated.

2. Assessment of Impact of Development Activities on Vegetation and Wil

2.1 Steps of the assessment



2.2 Baseline data collection

- **Secondary data** Forest Department records, Botanical Survey of India (BSI), Zoological Survey of India (ZSI), Wildlife Institute of India, satellite images and published literature.
- **Primary data** field surveys in different seasons.
- **Vegetation survey methods** quadrat method (1 m x 1 m for herbs, 5 m x 5 m for shrubs, 10 m x 10 m or 20 m x 20 m for trees), line transect and point-centred quarter method.
- **Wildlife survey methods** line transect and direct sighting, pug marks and scat (dung) counts, camera traps, bird point counts, nets for small animals, interviews with local people.
- **Aquatic life** plankton net, benthic sampler and fish catch.

2.3 Parameters studied

Parameter	Meaning
Density	Number of individuals per unit area
Frequency	Percentage of quadrats in which a species occurs
Abundance	Number of individuals per quadrat of occurrence
Basal area / Biomass	Size and weight of vegetation
Species diversity	Shannon-Wiener index H = - sum of (pi x ln pi)
Importance Value Index (IVI)	Relative density + relative frequency + relative dominance

- Also recorded: endemic, rare, endangered and threatened (RET) species as per IUCN Red List, Schedules of Wild Life Protection Act and BSI/ZSI lists.
- Note nearby protected areas such as national parks, sanctuaries, biosphere reserves, elephant corridors and Ramsar wetlands.

2.4 Types of impact

- **Direct impacts** clearing of vegetation, loss of habitat, killing of animals by vehicles and blasting, drowning by reservoirs.

- **Indirect impacts** air and water pollution reduce growth; dust on leaves reduces photosynthesis; noise and light disturb animals; increased human access leads to poaching and fire.
- **Habitat fragmentation** roads, canals and railway lines divide the forest into small patches; animals cannot move, breed or feed properly; gene flow reduces.
- **Edge effect** change in conditions at the edge of a cut forest.
- **Barrier effect** dams stop fish migration; fences stop animal movement.
- Introduction of invasive alien species (e.g. Lantana, Eichhornia - water hyacinth).
- Cumulative impacts from many projects in the same region.

Activity wise impacts

Activity	Impact on flora and fauna
Dams and reservoirs	Submergence of forest, fish migration blocked, downstream flow change
Mining	Loss of forest, dust, overburden dumps, water pollution
Roads and railways	Fragmentation, road kills, noise
Thermal power plant	Fly ash, SO2 damage to plants, thermal discharge
Industries	Toxic effluents, bioaccumulation in food chain
Urbanisation	Loss of habitat, light and noise pollution

2.5 Prediction and evaluation

- Predict the area of vegetation lost, number of trees to be felled, change in species diversity and loss of habitat for key species.
- **Methods** GIS and remote sensing for land cover change, habitat suitability models, population viability analysis, expert judgement and analogy with similar projects.
- **Significance depends on** rarity of species, ecological value, size of the area lost, whether it can be restored, and legal protection.

- Higher importance is given to schedule I species (tiger, elephant, lion), endemic species, breeding sites and corridors.

2.6 Mitigation measures

- Avoid sensitive areas by changing the site or alignment.
- Minimise clearing; use existing roads and degraded land.
- Compensatory afforestation (twice the degraded land area or equal area of non-forest land as per Forest Conservation Act 1980).
- Green belt of native species around the project.
- Wildlife corridors, underpasses and overpasses on highways, fish ladders at dams.
- Translocation of rare plants and animals; seed banks and nurseries.
- Restrict noise, lights and construction at breeding seasons.
- Strict control on poaching, fire and grazing; awareness for workers and villagers.
- Wildlife conservation plan and monitoring, budget given in the EMP.

3. Environmental Impact of Deforestation

Deforestation means the permanent removal of forest cover and its conversion to other uses such as agriculture, mining, industry, roads or settlement.

- India's forest cover is about 21 percent of the geographical area (national policy target is 33 percent).
- Tropical forests are lost mainly for agriculture, timber and development projects.

3.1 Causes of deforestation

- **Agricultural expansion** shifting (jhum) cultivation and conversion of forest to farm land.
- **Mining and quarrying** open cast mines destroy large areas.

- **River valley projects** dams submerge forests.
- **Industrialisation and urbanisation** factories, townships, roads.
- Fuel wood and fodder collection by rural people.
- Commercial logging for timber, paper and plywood.
- Overgrazing by cattle.
- Forest fires, natural or man made.
- Encroachment and illegal felling.
- **Infrastructure** roads, railways, power lines, defence needs.

3.2 Effects of deforestation

Area	Effect
Soil	Erosion, loss of fertility, landslides, siltation of rivers and reservoirs
Water	Floods in rainy season, drought in dry season, lower groundwater recharge, poor water quality
Climate	Less rainfall, rise in local temperature, more CO2 (global warming), loss of carbon sink
Biodiversity	Loss of habitat, extinction of species, human-animal conflict
Economy	Loss of timber, non timber forest products, medicinal plants, tourism
Society	Displacement and loss of livelihood of tribal people

- Desertification may occur in dry regions.
- Loss of the gene pool that could be useful for medicine and agriculture.
- Rise in air pollution as trees no longer filter dust and gases.

3.3 Control measures

- Afforestation, reforestation and social forestry.
- Strict implementation of Forest Conservation Act and Wild Life Act.
- Use of alternative fuels such as LPG and biogas; fuel efficient stoves.
- Sustainable forest management and Joint Forest Management (JFM) with local communities.

- Controlled grazing, fire prevention and watch towers.
- Use of recycled paper; agro-forestry and wasteland development.
- Remote sensing for monitoring forest cover.

Remember Chipko movement (1973, Uttarakhand, led by Sunderlal Bahuguna and Gaura Devi) is a famous people's movement to save trees.

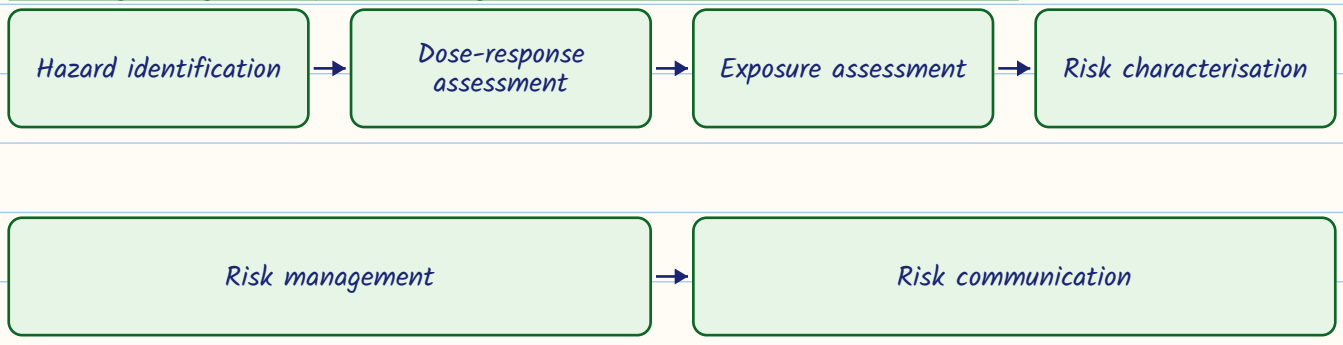
4. Risk Assessment and Treatment of Uncertainty

4.1 Introduction

- **Risk** the chance (probability) that a harmful event will occur, multiplied by how serious its consequence is. Risk = Probability x Consequence.
- Hazard is the source of danger (a toxic chemical, a fire); risk is the chance that the hazard actually harms someone.
- Environmental Risk Assessment (ERA) is the scientific process of estimating the likelihood and the size of adverse effects on human health and ecosystem because of exposure to hazards.
- ERA is used for hazardous industries, chemical storage, waste disposal sites, pipelines and accidents like Bhopal gas tragedy (1984).

Types of risk health risk, ecological risk, accident (safety) risk and natural hazard risk.

4.2 Key stages in performing environmental risk assessment



- **Stage 1 – Hazard identification** find the chemicals, processes and events that

may cause harm and what type of harm (toxic, carcinogenic, fire, explosion). Tools: HAZOP, FMEA, fault tree and event tree analysis, checklists.

- **Stage 2 – Dose-response (toxicity) assessment** relation between the amount of exposure (dose) and the effect on the organism. Uses LD50, LC50, NOAEL and reference dose (RfD) from lab and epidemiological data.
- **Stage 3 – Exposure assessment** who is exposed, by which route (inhalation, ingestion, skin contact), how much, how often and for how long. Uses dispersion models and food chain models.
- **Stage 4 – Risk characterisation** combining the above to give a numerical value of risk, for example excess cancer cases per million people, or hazard quotient (HQ = exposure / reference dose; HQ above 1 means concern).
- **Stage 5 – Risk management** decide what is acceptable and apply controls – engineering controls, safety distances, emergency plans, insurance.
- **Stage 6 – Risk communication** tell the public and authorities about the risk honestly and clearly.

Acceptable risk levels

- A lifetime cancer risk of 1 in a million (10 to the power minus 6) is generally considered negligible; up to 1 in 10,000 may be tolerated in some cases.
- Individual risk and societal risk (F-N curve) are both considered for major hazards.

4.3 Consequence analysis (for accidents)

- **Possible accidents** toxic gas release, fire, pool fire, jet fire, vapour cloud explosion, BLEVE.
- Models predict the distance up to which heat radiation, overpressure or toxic concentration crosses a harmful limit.
- Output is a map of damage zones used for the on-site and off-site emergency plan, siting and buffer zones.

- Disaster Management Plan (DMP) includes alarm system, evacuation, first aid, fire fighting and communication.

4.4 Treatment of uncertainty

Uncertainty means that predictions are not exactly known because of missing data, simple models or natural variation.

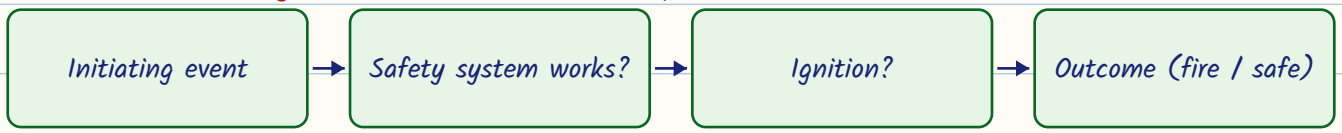
- **Sources** errors in measurement and sampling, variability among people and species, model assumptions, extrapolation from animals to humans or from high dose to low dose, and incomplete knowledge.
- **Types** parameter uncertainty, model uncertainty and scenario uncertainty. Variability (real differences) is different from uncertainty (lack of knowledge).
- **Ways to treat it**
  - Sensitivity analysis - change one input at a time and see its effect on the result.
  - Monte Carlo simulation - run the model thousands of times with random inputs from probability distributions and get a range of results.
  - Use of safety (uncertainty) factors, usually 10 each for animal to human and for human to human variation.
  - Worst case and best case scenarios; conservative assumptions.
  - Confidence intervals and probability statements.
  - Collect more data, peer review and expert opinion.
  - Adaptive management - monitor and adjust after the project starts.

Key idea uncertainty cannot be removed completely, but it should be identified, quantified and clearly shown to decision makers.

4.5 Hazard identification tools in brief

- **HAZOP (Hazard and Operability study)** a team examines each part of a process using guide words such as NO, MORE, LESS, REVERSE, OTHER THAN, and asks what deviation can occur, its causes, consequences and safeguards.

- **FMEA (Failure Mode and Effect Analysis)** lists every component, how it can fail and what effect the failure has.
- **Fault tree analysis (FTA)** a top-down diagram that starts from an undesired top event (for example, tank rupture) and finds all the combinations of causes using AND and OR gates.
- **Event tree analysis (ETA)** a bottom-up diagram that starts from an initiating event (for example, gas leak) and follows the possible outcomes (ignition or no ignition, safety system working or failing).
- **What-if analysis and checklists** simple tools for small units.



4.6 Simple risk numerical

A chemical plant has a probability of tank failure of 0.002 per year. If it happens, the expected number of deaths is 50.

- Risk = Probability x Consequence
- Risk = 0.002 x 50 = 0.1 deaths per year

Result expected loss is 0.1 deaths per year. If the company installs a safety valve that reduces the probability to 0.0004, then the new risk is 0.02 deaths per year, a fall of 80 percent. This shows how risk assessment helps in judging whether a safety investment is worth it.

4.7 Hazard quotient example

- A pollutant has a reference dose RfD = 0.02 mg/kg/day.
- Estimated exposure of people through drinking water = 0.05 mg/kg/day.
- HQ = Exposure / RfD = 0.05 / 0.02 = 2.5
- Since HQ is greater than 1, there is a health concern and the source must be controlled.

4.8 Ecological risk assessment

- It estimates the risk to plants, animals and ecosystems from stressors such as chemicals, habitat loss or invasive species.
- **Steps** problem formulation, analysis (exposure and effects) and risk characterisation.
- Uses indicator species, bioassays (LC50 on fish and Daphnia), bioaccumulation and biomagnification data.
- **Bioaccumulation** build up of a chemical in an organism. Biomagnification :: increase in concentration at each higher level of the food chain (for example DDT and mercury).

5. Advantages of Environmental Risk Assessment

- Gives a scientific and logical basis for decisions.
- Helps in identifying the most serious hazards so that money is spent where it is most useful.
- Allows comparison of different options and alternatives.
- Helps in setting standards and priorities.
- Reduces chances of accidents, health damage and loss of property.
- Helps in preparing emergency and disaster management plans.
- Increases public confidence and transparency through risk communication.
- Helps in legal compliance and in obtaining clearances and insurance.
- Encourages cleaner technology and safer design.

Limitations

- Needs a lot of data and skilled persons.
- Results depend on assumptions and are uncertain.
- Hard to measure ecological and long term effects.
- Public may not understand numerical risk values.

5A. Biodiversity Conservation in Projects

- **In situ conservation** protecting species in their natural habitat - national parks, wildlife sanctuaries, biosphere reserves, sacred groves.
- **Ex situ conservation** outside the habitat - zoos, botanical gardens, gene banks, seed banks, cryopreservation.
- **Protected area categories in India** national parks (no human activity allowed), wildlife sanctuaries (limited activity), conservation reserves, community reserves and biosphere reserves.
- Eco-sensitive zones (ESZ) around protected areas, usually up to 10 km, where industries and mining are regulated.
- **Species protection** IUCN Red List categories are Extinct, Critically Endangered, Endangered, Vulnerable, Near Threatened and Least Concern.
- Project proponent must prepare a Wildlife Conservation Plan with the State Wildlife Warden when a Schedule I species is present within the study area.

Case Example - Silent Valley (Kerala)

- A hydroelectric project proposed on the Kunthipuzha river in the 1970s threatened the evergreen rainforest and the lion tailed macaque.
- Strong protests by scientists and the public stopped the project; the area was declared a National Park in 1984.
- **Lesson** ecological value can be more important than short term power benefits; EIA should include biological studies at the very start.

Case Example - Sardar Sarovar and Tehri dams

- Submergence of forests and agricultural land and displacement of thousands of families.
- Shows the link between biological, social and cultural impacts; resettlement and rehabilitation (R&R) plans must be part of the EIA.

6. Case Example – Bhopal Gas Tragedy (1984)

- Leak of methyl isocyanate (MIC) from the Union Carbide pesticide plant on the night of 2-3 December 1984.
- Thousands of people died and lakhs were affected; long term health and soil and groundwater contamination followed.
- **Causes** storage of large quantity of MIC, poor maintenance, failure of safety systems, plant near a densely populated area, no proper emergency plan.
- **Lessons** need for hazard analysis, safe siting, buffer zones, emergency preparedness and public awareness. Led to the Environment Protection Act 1986 and the Public Liability Insurance Act 1991.

7. Key Points to Remember

- Vegetation is studied by quadrats and transects; fauna by transects, pug marks and camera traps.
- Diversity is measured by the Shannon-Wiener index.
- **Main causes of deforestation** agriculture, mining, dams, industries, fuel wood, logging.
- Risk = probability x consequence.
- **ERA stages** hazard identification, dose-response, exposure assessment, risk characterisation, risk management and communication.
- Uncertainty is treated by sensitivity analysis, Monte Carlo simulation and safety factors.

Important questions (1) Explain the impact of development activities on vegetation and wildlife. (2) Causes and effects of deforestation. (3) Explain the key stages in environmental risk assessment. (4) How is uncertainty treated? (5) Advantages of environmental risk assessment.

Subject: ENVIRONMENTAL IMPACT ASSESSMENT

Subject Code: 23A01504c

UNIT - IV

Environmental Audit

I. Introduction

An environmental audit is a systematic, documented, periodic and objective evaluation of how well an organisation, its management and its equipment are performing in protecting the environment.

Definition (ICC, 1989) "A management tool comprising a systematic, documented, periodic and objective evaluation of how well environmental organisation, management and equipment are performing, with the aim of helping to safeguard the environment by facilitating management control of environmental practices and assessing compliance with company policies, including regulatory requirements and standards."

- **In simple words** an environmental audit is a health check-up of a company's environmental performance.
- It is done on existing activities, while EIA is done before a project starts.
- Audit is a tool for management; it checks compliance, finds weaknesses and suggests improvements.
- In India, Environmental Statement (Form V) must be submitted every year by 30 September to the State Pollution Control Board, under the Environment (Protection) Rules, 1992 (introduced in 1993). This is a statutory form of environmental audit.

Audit versus EIA

Point	EIA	Environmental audit
Time	Before project	During running of project

Purpose	Predict impact	Verify actual performance and compliance
Basis	Predictions and models	Records, site inspection, measurements
Result	Clearance decision	Audit report and action plan

2. Environmental Audit and Environmental Legislation

- Laws set standards for emission, effluent, noise, waste and safety. Audit checks whether the industry follows them.
- **Main legislation covered** Water Act 1974, Air Act 1981, Environment Protection Act 1986, Hazardous Waste Rules, Noise Pollution Rules 2000, Biomedical Waste Rules, Solid Waste Rules and conditions of the Consent to Establish and Consent to Operate.
- **Audit also checks** EIA clearance conditions, EMP implementation, and ISO 14001 requirements.
- Failure to comply may lead to fines, closure notices, prosecution and loss of reputation.
- A regular audit helps the industry to find and correct violations before the authorities do.
- Government has also made audit mandatory in different forms, such as the Environmental Statement, hazardous waste returns and the Energy audit under the Energy Conservation Act 2001.
- Third party audit by recognised auditors is encouraged by the pollution control boards and in the Charter on Corporate Responsibility for Environmental Protection (CREP).

3. Objectives of Environmental Audit

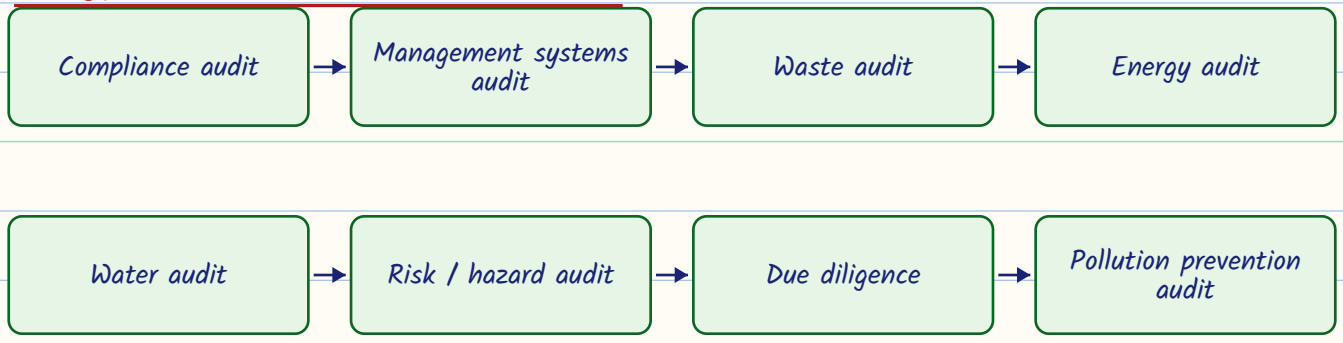
- To check compliance with laws, standards and permit conditions.
- To assess the effectiveness of pollution control equipment and the

environmental management system.

- To identify environmental risks and liabilities early.
- To find chances for saving raw materials, water, energy and money (waste minimisation and cleaner production).
- To improve environmental awareness and training among employees.
- To check that the company's own environmental policy is being followed.
- To provide reliable information to management, the public and regulators.
- To improve the public image of the organisation and relationship with the community.
- To help in insurance, loans and in buying or selling of properties (due diligence).
- To support continual improvement.

Benefits cost saving, legal safety, better image, fewer accidents, better resource use, better management of waste.

4. Types of Environmental Audit



- **Compliance audit** checks whether the plant follows all laws, rules, permits and standards. It is the most common type.
- **Environmental Management System (EMS) audit** checks whether the system (policy, planning, implementation, checking, review) as per ISO 14001 is working.

- **Waste (and emission) audit** studies the sources, types and quantities of waste, and ways for reduction, reuse and recycling.
- **Energy audit** studies the energy use and finds saving options.
- **Water audit** studies the water balance, losses, reuse and recycling.
- **Risk or hazard audit** studies the chances of accidents and the safety of storage, handling and emergency plans.
- **Due diligence (transactional) audit** done before purchase, merger or sale of a company or land to find hidden environmental liabilities.
- **Pollution prevention (minimisation) audit** finds ways to reduce pollution at the source.
- **Health and safety audit** checks occupational health and safety systems.
- **Product / life cycle audit** studies environmental effect of a product from raw material to disposal.

Based on who does it

- **Internal audit (first party)** done by the company's own trained staff.
- **External audit (second party)** done by customers or buyers.
- **Third party audit** done by independent outside auditors or certifying bodies. More credible.

Based on scope

- Site audit, corporate audit, issue specific audit, and follow up audit.

5. Audit Protocol

Audit protocol is a written set of procedures, checklists and questions, prepared beforehand, which guides the audit team on what to examine and how to record findings.

- It gives a uniform and consistent approach, and makes sure no important area is missed.

- It is used as a working document and its filled form becomes the evidence.

Contents of a protocol

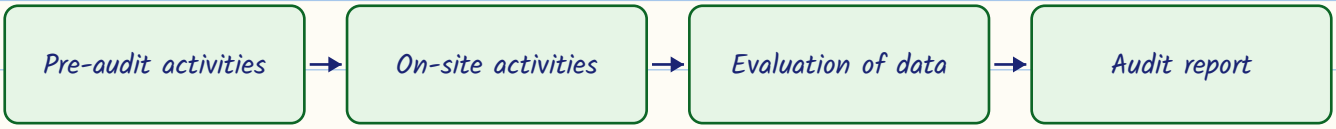
- Audit objective, scope and criteria (which laws and standards).
- **List of areas to be audited** raw material storage, process, utilities, air emissions, wastewater, solid and hazardous waste, noise, safety and emergency, records.
- Checklists and questionnaires for each area.
- Sampling and measurement methods.
- **Documents to be reviewed** consents, returns, monitoring reports, maintenance records, licences.
- Interview list of persons to meet.
- Format for recording observations and for the report.
- Rules of confidentiality and ethics.

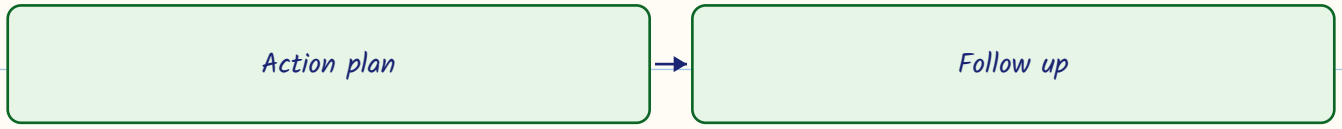
Sample questions in a checklist

- Is a valid Consent to Operate available? Are the limits followed?
- Is the effluent treatment plant working and are records kept daily?
- Are hazardous wastes labelled, stored and sent only to authorised recyclers?
- Are stack heights and monitoring facilities as per norms?
- Are employees trained for emergency?

6. Stages of Environmental Audit

The audit is carried out in three main stages: Pre-audit, Onsite audit (activities) and Post-audit.





6.1 Pre-audit (planning) activities

- Decide the objective, scope, depth and time of the audit.
- Select the audit team and the team leader; make sure they are independent and qualified.
- Inform the management and the plant personnel and fix the schedule.
- **Collect background information** site layout, process flow diagram, products, raw materials, past reports, licences, previous audit findings, complaints.
- Prepare the audit protocol, checklists and questionnaire.
- Plan the sampling, instruments and resources.
- Hold a preliminary meeting with the site managers.

6.2 Onsite activities

- Opening meeting with plant management to explain plan and procedure.
- **Understanding the management system and controls** review of the policy, responsibilities, training and procedures (this is checked with questionnaires and interviews).
- Assessing strengths and weaknesses of the internal controls.
- Physical inspection (site walk through) of the process, storage, pollution control units, drains, waste yard and housekeeping.
- **Collecting audit evidence** observation, interviews, document and record checks, sampling and measurements.
- **Verifying the evidence** cross checking and confirming findings with more than one source.
- Evaluating the findings against the criteria (laws, standards, company policy).
- Closing meeting, in which preliminary findings are discussed with management.

Evidence types physical (observation and samples), documentary (records and reports), testimonial (statements from persons) and analytical (laboratory results and calculations).

6.3 Post-audit activities

- Evaluation of audit data and preparation of the audit report.
- Management response and action plan with time schedule and responsibility.
- Follow up audit to check that the corrective actions have been carried out.

7. Evaluation of Audit Data

- All data from checklists, observations, measurements and records are collected and organised area wise.
- The findings are compared with the audit criteria – legal limits, permit conditions and company standards.
- **Each finding is classified** compliance, non-compliance (major or minor), or an observation / opportunity for improvement.
- Cause of each non-compliance is found, for example lack of training, equipment failure or poor maintenance.
- Findings are checked by different sources to be sure they are correct and fair.
- Risk ranking is done by how serious the effect is and how likely it is to occur, so that priority is given to the most severe items.
- Measured values (stack emissions, effluent quality, noise) are compared with standards in a table.
- Material balance and water balance are checked to find losses and wastage.
- Trends are studied over time (for example, last three years of monitoring data).
- Good practices are also noted and appreciated.

Finding	Criteria	Status	Priority

Effluent BOD 45 mg/L	Limit 30 mg/L	Non-compliance	High
Stack PM 120 mg/Nm3	Limit 150	Compliant	-
Hazardous waste not labelled	Rules	Minor non-compliance	Medium
Water meter missing	Good practice	Observation	Low

8. Preparation of the Audit Report

The audit report is the final document that presents the findings, conclusions and recommendations to the management.

Qualities of a good report

- Clear, accurate, brief, objective, balanced and complete.
- Supported by evidence; written in simple language.
- Constructive; points out good practices and not only the faults.
- Submitted on time and kept confidential.

Format of the report

- Title page and contents.
- Executive summary.
- **Introduction** objective, scope, criteria, dates, team and the methodology.
- Description of the facility and its activities.
- **Audit findings** area wise strengths, non-compliances and observations.
- Conclusions.
- Recommendations and corrective action plan, with priority, responsible person, time and cost.
- **Annexures** checklists, test results, photographs, list of persons met and documents examined.
- Signatures of the audit team.
- The draft report is discussed with the management for factual accuracy before the final report is issued.

Note the audit report is the property of the organisation. Findings should be used to improve, not to blame.

9. Environmental Statement (Form V) – Key Points

- Submitted by every industry requiring consent under the Water or Air Act, each year before 30th September for the financial year ended 31st March.
- **Contents** water and raw material consumption, pollutants discharged, hazardous waste generated, solid waste, steps taken for pollution control, investment, and proposals for further improvement.
- It is a self audit report to the State Pollution Control Board.

9A. Qualities and Duties of an Environmental Auditor

- Technical knowledge of the process, pollution control and environmental laws.
- Independence and honesty; no conflict of interest.
- Good observation, questioning and listening skills.
- Ability to write clear reports and communicate tactfully.
- Maintains confidentiality of company information.
- Uses a team approach: process engineer, chemist or analyst, safety expert and legal expert.
- **Duties** plan the audit, follow the protocol, collect and verify evidence, report truthfully and follow up.

9B. Specific Audits in Detail

Waste audit (step by step)

- **Step 1** draw the process flow diagram and mark every input and output.
- **Step 2** measure the quantity and quality of each waste stream – solid, liquid, gas and hazardous.

- **Step 3** prepare a material balance - Input = Product + Waste + Accumulation.
- **Step 4** find the sources of highest waste and the cost of disposal.
- **Step 5** suggest options - source reduction, reuse, recycling, recovery, treatment.
- **Step 6** estimate the cost and the payback period of each option and select the best.

Waste hierarchy Reduce at source -> Reuse -> Recycle -> Recover energy -> Treat -> Dispose safely (the last option).

Water audit

- Collect data on water sources, meters, uses in process, cooling, boiler, domestic and gardening.
- Prepare a water balance diagram: Total intake = Consumption + Losses + Wastewater discharged.
- Find leakages and overuse; check possibilities of recycle and reuse of treated water and rain water harvesting.
- **Example** A plant takes 1000 kL/day. Process uses 600, cooling loses 150 by evaporation, domestic 50 and wastewater discharged is 180. Total accounted = 980. Unaccounted loss = 20 kL/day (2 percent), probably leakage, which must be traced.

Energy audit

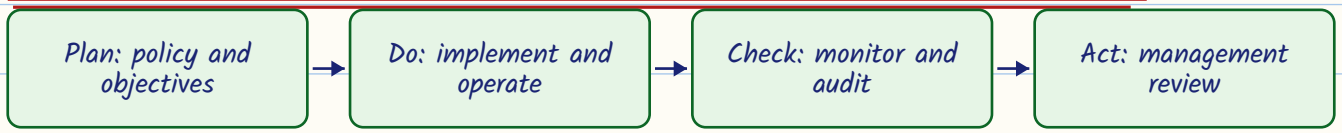
- Preliminary audit (walk through, using bills) and detailed audit (measurements with instruments).
- Study of power factor, motors, lighting, boilers, compressed air and heat recovery.
- Result is a list of energy saving measures with investment and payback.

Hazardous waste and safety audit

- Checks storage, labelling, manifest system, authorised transporters and disposal sites.

- Checks fire fighting, alarms, personal protective equipment, safety training, mock drills and the on-site emergency plan.

9C. ISO 14001 Based EMS Audit (Plan – Do – Check – Act)



- The auditor checks that the policy is approved by top management and known to all workers.
- Checks that significant environmental aspects and legal requirements are identified.
- Checks training, communication, document control and emergency preparedness.
- Checks monitoring records, calibration of instruments, non-conformity handling and internal audits.
- Management review must be done at planned intervals to give continual improvement.

9D. Advantages and Limitations of Environmental Audit

Advantages

- Early warning of problems and compliance gaps.
- Saves money by reducing waste, energy and water use.
- Improves safety and reduces accidents and liability.
- Builds confidence among regulators, banks, insurers and customers.
- Encourages employee participation and responsibility.
- Gives a base for ISO 14001 certification and for green marketing.

Limitations

- Needs time, money and trained persons.
- If done carelessly it becomes only a paper exercise.
- Fear of exposing violations may cause managers to hide information.
- Results depend on the quality of records and honesty of staff.
- Without follow up, the recommendations may not be implemented.

9E. Green Audit and Other Related Terms

- **Green audit** environmental audit of institutions like colleges, hospitals and offices - covers energy, water, waste, greenery and transport.
- **Social audit** checks the social effects of a project on workers and community.
- **Environmental performance evaluation (EPE)** uses indicators such as emission per tonne of product, water per unit and waste generated.
- **Life Cycle Assessment (LCA)** studies the environmental load of a product from cradle to grave.
- **Eco-labelling (Ecomark)** label for environment friendly products in India.

10. Key Points to Remember

- Environmental audit = systematic, documented, periodic and objective evaluation of environmental performance.
- **Types** compliance, EMS, waste, energy, water, risk, due diligence, pollution prevention; internal, external, third party.
- Protocol = checklists and procedures that guide the audit.
- Stages = pre-audit, onsite audit, post audit.
- **Onsite steps** understand system -> assess controls -> gather evidence -> evaluate -> report.
- Audit follow up ensures that recommendations are really implemented.

Important questions (1) Define environmental audit and explain its objectives. (2) Types of environmental audit. (3) What is an audit protocol? (4) Explain the stages of an environmental audit. (5) How is audit data evaluated and the report prepared?

Subject: ENVIRONMENTAL IMPACT ASSESSMENT

Subject Code: 23A01504c

UNIT – V

Environmental Acts and Notifications

I. Introduction – Environmental Legislation and its Objectives

Environmental legislation means the laws made by the government to protect the environment, control pollution and conserve natural resources.

- **Constitutional basis** Article 48A (Directive Principle) – the State shall protect and improve the environment and safeguard forests and wildlife. Article 51A(g) (Fundamental Duty) – every citizen shall protect and improve the natural environment. Article 21 gives the right to a clean environment (Supreme Court judgments).
- Stockholm Conference 1972 on the Human Environment led India to make many environmental laws.

Objectives of environmental legislation

- To prevent, control and abate pollution of air, water, soil and noise.
- To conserve forests, wildlife and biodiversity.
- To fix standards for emission and effluent discharge.
- To give powers to authorities to inspect, monitor and punish violations.
- To ensure sustainable use of natural resources.
- To make industries responsible (polluter pays principle) and to promote public participation.
- To bring EIA and environmental audit under legal control.

Year	Law
1972	Wild Life (Protection) Act
1974	Water (Prevention and Control of Pollution) Act

1980	Forest (Conservation) Act
1981	Air (Prevention and Control of Pollution) Act
1986	Environment (Protection) Act
1991	Public Liability Insurance Act
2002	Biological Diversity Act
2010	National Green Tribunal Act

2. The Environment (Protection) Act, 1986

- Enacted after the Bhopal gas tragedy (1984), under Article 253 of the Constitution to implement the decisions of the Stockholm Conference. Came into force on 19 November 1986.
- It is an umbrella (framework) Act that fills the gaps in the earlier Acts and gives wide powers to the Central Government.
- Has 26 sections divided into 4 chapters.
- **Definitions (Section 2)** environment includes water, air, land and their inter-relationship with human beings, other living creatures, plants, micro-organisms and property. Environmental pollutant is any solid, liquid or gaseous substance present in such concentration as may be injurious to the environment.

Important provisions

- **Section 3** Central Government can take all measures for protecting and improving the environment, including setting up authorities, fixing standards, restricting industrial locations and laying down procedures for safeguards against accidents.
- **Section 5** power to issue directions, including closure, prohibition or regulation of any industry, and stopping electricity and water supply.
- **Section 6** power to make rules for standards of quality, maximum limits of

pollutants, handling of hazardous substances, restricted areas and so on.

- **Section 7** no person shall emit or discharge pollutants above the prescribed standards.
- **Section 8** hazardous substances must be handled as per procedures and safeguards.
- **Section 10** powers of entry and inspection; Section 11 :: power to take samples.
- **Section 15** penalty for contravention - imprisonment up to 5 years, or fine up to Rs. 1 lakh, or both; continuing violation extra fine of Rs. 5000 per day; after one year of contravention the imprisonment may extend to 7 years.
- **Section 19** court can take notice of a complaint made by the Government, or by any person after giving 60 days notice.
- **Section 17** offences by Government departments and Section 16 :: offences by companies (the person in charge is responsible).

Rules made under the Act

- **Environment (Protection) Rules 1986** standards for emission and effluent.
- EIA Notification 1994 (replaced in 2006), Hazardous Waste Rules, Biomedical Waste Rules, Noise Pollution Rules 2000, Coastal Regulation Zone Notification, Environmental Statement (1993).

3. The Water (Prevention and Control of Pollution) Act, 1974

- **Objective** prevention and control of water pollution and maintaining or restoring the wholesomeness of water. It is the first major law on pollution in India.
- **Water cess Act 1977** tax on industries using water, used to finance the Boards.
- Water pollution means contamination of water or change in its properties, or discharge of any sewage, trade effluent or other substance which is likely to

cause a nuisance or harm to public health, animals, plants or aquatic life.

Institutions

- **Central Pollution Control Board (CPCB)** formed in 1974; advises the Central Government, coordinates State Boards, sets national standards, collects data and runs training.
- **State Pollution Control Boards (SPCBs)** plan programmes, inspect industries, give consent, lay down effluent standards, monitor and take action.
- Board has a chairman, members from government, local bodies, farming, fishing and industry, and a member secretary.

Main provisions

- **Section 24** no person shall knowingly cause or permit any poisonous, noxious or polluting matter to enter a stream, well, sewer or on land.
- **Section 25** consent (permission) of the State Board is required before starting any industry or treatment system that may discharge effluent - Consent to Establish (CTE) and Consent to Operate (CTO).
- **Section 26** existing industries must also obtain consent; Section 27 :: Board can refuse or give consent with conditions.
- **Sections 20-23** powers to take samples, enter and inspect.
- **Section 33A** power to issue directions, including closure of the industry and stopping of power and water.
- **Penalties** Sections 41 to 44; imprisonment from 1 year 6 months up to 6 years with fine for violations (as amended in 1988). Section 47 :: offences by companies.
- Appeals go to the Appellate Authority / National Green Tribunal.

4. The Air (Prevention and Control of Pollution) Act, 1981

- Enacted to implement the decisions of the Stockholm Conference and to

prevent, control and reduce air pollution. Noise was added as an air pollutant by the 1987 amendment.

- Air pollutant means any solid, liquid or gaseous substance (including noise) present in the atmosphere in such concentration as may be harmful to human beings, living creatures, plants, property or the environment.
- Implemented by the CPCB and State Boards, which have the same structure as under the Water Act.

Main provisions

- **Section 16** functions of the CPCB - to advise the Government, plan nation wide programmes, lay down standards for air quality and provide technical help.
- **Section 17** functions of the State Boards - plan programme, inspect, lay down emission standards for industry and automobiles.
- **Section 19** State Government can declare Air Pollution Control Areas (after consulting the Board) and prohibit use of certain fuels and appliances.
- **Section 21** consent from the State Board is required to establish and operate any industrial plant in a pollution control area.
- **Section 22** no person shall emit air pollutants in excess of the standards.
- **Section 20** control of emissions from automobiles.
- **Section 31A** power to give directions including closure.
- **Section 37** penalty for failure to comply - imprisonment from 1 year 6 months up to 6 years with fine; continuing offence additional fine per day.
- **Section 40** offences by companies.

National Ambient Air Quality Standards (NAAQS 2009) limits for 12 pollutants including SO2, NO2, PM10, PM2.5, ozone, lead, CO, ammonia, benzene, benzo(a)pyrene, arsenic and nickel.

5. The Wild Life (Protection) Act, 1972

- **Objective** to protect wild animals, birds and plants, to control hunting, poaching and trade in wildlife products, and to protect their habitats.
- Gives the legal basis for national parks, sanctuaries and conservation areas. Amended in 1982, 1991, 2002, 2006, 2013 and 2022.
- **Authorities** Indian Board for Wildlife (National Board for Wildlife), Chief Wildlife Warden, wildlife wardens, National Tiger Conservation Authority (since 2006 amendment) and Wildlife Crime Control Bureau.

Main provisions

- Hunting of wild animals in Schedules I–IV is prohibited, except in special cases (danger to human life or disease) with permission.
- **Protected areas** Sanctuaries (limited human activity), National Parks (no grazing or private rights), Conservation Reserves, Community Reserves and Tiger Reserves.
- **Schedules** Schedule I and part II of Schedule II give absolute protection with highest penalty (for example tiger, elephant, lion, rhino, peacock). Schedules III and IV protect less endangered species. Schedule V is for vermin. Schedule VI is for protected plants (for example Kuth, Pitcher plant). (Rationalised in the 2022 amendment.)
- Trade in animal articles, trophies and ivory is banned or licensed.
- Zoos have to be recognised by the Central Zoo Authority.
- **Penalty** for Schedule I offences, imprisonment 3 to 7 years and fine of at least Rs. 10,000 (higher later), more for repeat offences.
- Wildlife areas cannot be changed or de-notified without approval of the legislature and the National Board for Wildlife. The Act is important in EIA because projects near protected areas need Board permission.

Related Act

- **Forest (Conservation) Act, 1980** prior approval of the Central Government is needed to use forest land for non forest purposes; compensatory afforestation is compulsory.

6. Provisions in the EIA Notification (14 September 2006)

- Issued by the Ministry of Environment, Forest and Climate Change (MoEFCC) under the Environment (Protection) Act 1986 and Rule 5(3) of the EP Rules 1986. It replaced the EIA Notification of 27 January 1994.
- Prior Environmental Clearance (EC) is required for new projects and for expansion or modernisation listed in the Schedule (about 39 types) - mining, thermal and nuclear power, river valley, ports, airports, highways, industrial estates, chemicals, cement, pharmaceuticals, townships and others.

Categories of projects

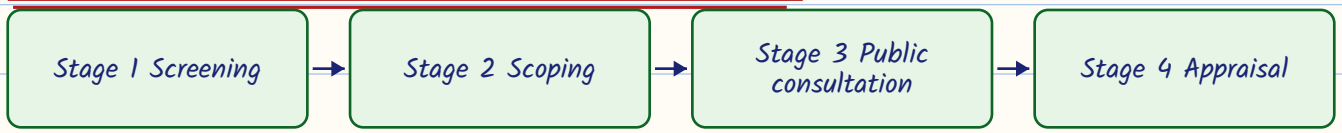
Category	Appraised by	Remarks
Category A	MoEFCC (Delhi) with Expert Appraisal Committee (EAC)	Large and highly polluting projects; EIA and public consultation compulsory
Category B1	State Environment Impact Assessment Authority (SEIAA) with State EAC (SEAC)	EIA report required
Category B2	SEIAA with SEAC	Smaller projects; no EIA report, only Form 1 and conceptual plan

- Category depends on the size, capacity, location (inter-state borders, protected areas, eco-sensitive zones) and type of the project; general conditions apply.
- Expert Appraisal Committees consist of experts in various fields such as environmental science, ecology, engineering and law.
- Some projects do not need public hearing, such as expansion of existing units

without more land and modernisation.

- The validity of the clearance depends on the project (for example 10 years for river valley, 30 years for mining, 5 years for others, later changes).
- Clearance conditions must be followed, and the proponent has to submit half yearly compliance reports.

7. Procedure for Environmental Clearance



Stage 1 – Screening (only for Category B)

- The proponent submits Form 1 (application) with a conceptual plan.
- SEAC decides whether the project is B1 (EIA needed) or B2 (no EIA).

Stage 2 – Scoping

- The EAC / SEAC studies the application and finalises Terms of Reference (ToR) for the EIA report, normally within 60 days.
- Proponent may suggest the ToR; the ToR is displayed on the website. If no reply in 60 days, the proposed ToR is deemed accepted.
- Baseline data collection and EIA study are done on the basis of the ToR.

Stage 3 – Public consultation

- Has two parts: (a) public hearing at the site or near it, and (b) written responses from interested persons.
- The State Pollution Control Board advertises the hearing in two newspapers (one in the local language) at least 30 days before; draft EIA is made available to the public.
- Public hearing is held within 45 days of the request; it is chaired by the District Magistrate or a representative.

- The proceedings are video recorded and sent to the regulatory authority; written comments are accepted for 30 days.
- The concerns of the public are addressed and the EIA report is finalised.

Stage 4 – Appraisal

- The final EIA report with the EMP and public hearing details is submitted to the authority.
- The EAC / SEAC examines it in detail and may ask for more data; appraisal is completed within 60 days.
- The authority takes a final decision (grant, grant with conditions or reject) within 45 days of the recommendation. The total period is 105 days.
- If no decision is communicated in the time, the proponent may presume deemed clearance, as per the rules.
- Grant of EC, with its conditions, is put on the website and in newspapers.



8. Procedure for Conducting the EIA Report

- The EIA study is done by an accredited consultant (as per NABET / QCI scheme).
- Contents as per Appendix III of the 2006 Notification and the ToR :
  - Executive summary and introduction; project description (need, location, size, process, raw materials, utilities, pollution load).
  - Description of the environment (baseline) - air, water, noise, soil, ecology, socio-economic - for one season (other than monsoon).
  - Anticipated environmental impacts and mitigation measures.
  - Analysis of alternatives (technology and site).

  - Environmental monitoring programme.
  - **Additional studies** risk assessment, disaster management plan, social impact assessment and R and R plan.
  - Project benefits.
  - Environmental Management Plan with budget.
  - Summary and conclusion; names of consultants.
- Data is collected by approved laboratories and the report is submitted to the authority and the Board.

9. Evaluation of the EIA Report

- Done by the Expert Appraisal Committee through a review that checks :
  - Whether the report follows the ToR and the proper format.
  - Is the baseline data adequate, recent and reliable?
  - Are all the impacts identified and are the prediction methods suitable?
  - Are the alternatives considered and the site suitable?
  - Are mitigation measures practical and enough? Is the EMP and the budget sufficient?
  - Have public concerns and hearing outcomes been addressed?
  - Does the project comply with laws, standards, and with forest and wildlife clearances?
  - Is the risk and disaster management plan adequate?
- The committee may ask the proponent to submit extra data or revise the report and may visit the site.
- **Final recommendation** approve, approve with conditions or reject.
- **Typical review criteria** completeness, accuracy, objectivity, clarity and usefulness for decision making.
- Public can also raise grievance with the National Green Tribunal (NGT) against

the clearance.

10. Evaluation of Audit Data and Preparation of Audit Report (Recap)

- Audit data (checklists, measurements, records and observations) are organised and compared with legal and company criteria.
- Findings are classed as compliance, non-compliance or observation, and ranked by priority.
- **The report contains** executive summary, introduction, facility description, findings, conclusions, recommendations with action plan and annexures.
- The draft is discussed with the management and then finalised.

11. Post Audit Activities

- Management reviews the report and prepares an action plan - what to do, who will do it, by when, and at what cost.
- **Corrective actions** repair or upgrade of pollution control equipment, training, better record keeping, safe storage of waste.
- **Preventive actions** changes in the process, cleaner production, replacing hazardous materials.
- Follow up audit or verification after a fixed period to check that the actions are completed.
- Reporting to the Board and to the public (Environmental Statement).
- Updating the environmental policy, objectives and the EMS; continual improvement.
- Keeping all documents for future reference.



12. Concept of ISO and ISO 14000

ISO

- ISO = International Organization for Standardization, formed in 1947 with headquarters in Geneva, Switzerland.
- It is a non-governmental federation of national standards bodies from more than 160 countries (India is represented by the Bureau of Indian Standards, BIS).
- **Its standards give uniform specifications for products, services and systems for quality, safety and efficiency. Examples** ISO 9000 (quality management), ISO 45001 (occupational health and safety) and ISO 14000 (environment).

ISO 14000 family

- A series of international standards on environmental management, first published in 1996 by ISO Technical Committee 207. It is a voluntary standard.
- **Purpose** to help organisations minimise harmful effects on the environment, comply with laws and improve continually.

Standard	Subject
ISO 14001	Environmental Management System – requirements (the only certifiable standard)
ISO 14004	Guidelines on EMS principles and support
ISO 14006	Eco-design
ISO 14020 series	Environmental labelling (eco-labels)
ISO 14031	Environmental performance evaluation
ISO 14040 / 14044	Life Cycle Assessment
ISO 14064	Greenhouse gas accounting and verification
ISO 19011	Guidelines for auditing management systems

ISO 14001 (EMS) – key elements

- Follows the Plan – Do – Check – Act cycle.
  - **Plan** environmental policy, identifying aspects and impacts, legal requirements, objectives and targets.
  - **Do** resources, roles, training, communication, documentation, operational control and emergency preparedness.
  - **Check** monitoring and measurement, evaluation of compliance, internal audit, nonconformity and corrective action.
  - **Act** management review and continual improvement.
- Latest version is ISO 14001:2015, which stresses leadership, risk based thinking and the life cycle perspective.

Benefits of ISO 14001

- Better compliance with laws and fewer penalties.
- Saving of raw material, energy and money; less waste.
- Improved image and trust of customers, banks and the public.
- Competitive advantage in the global market.
- Systematic management and continual improvement of environmental performance.
- Certification is given by accredited third party certification bodies after an audit and is valid for 3 years with yearly surveillance audits.

13. Key Points to Remember

- EP Act 1986 is the umbrella Act (Bhopal, Stockholm); Water Act 1974; Air Act 1981; Wild Life Act 1972.
- CPCB (1974) and SPCBs give Consent to Establish and Consent to Operate.
- **EIA Notification 2006** categories A, B1, B2; stages are screening, scoping, public consultation and appraisal.

- **Authorities** MoEFCC with EAC for category A; SEIAA with SEAC for category B.
- Post audit means action plan and follow up audit.
- ISO 14001 is the certifiable EMS standard in the ISO 14000 family.

Important questions (1) Explain the salient features of the Environment (Protection) Act 1986. (2) Water Act 1974 and Air Act 1981 provisions. (3) Wild Life Act 1972. (4) Explain the procedure for environmental clearance as per EIA Notification 2006. (5) How is an EIA report prepared and evaluated? (6) Post audit activities. (7) Write about ISO 14000.